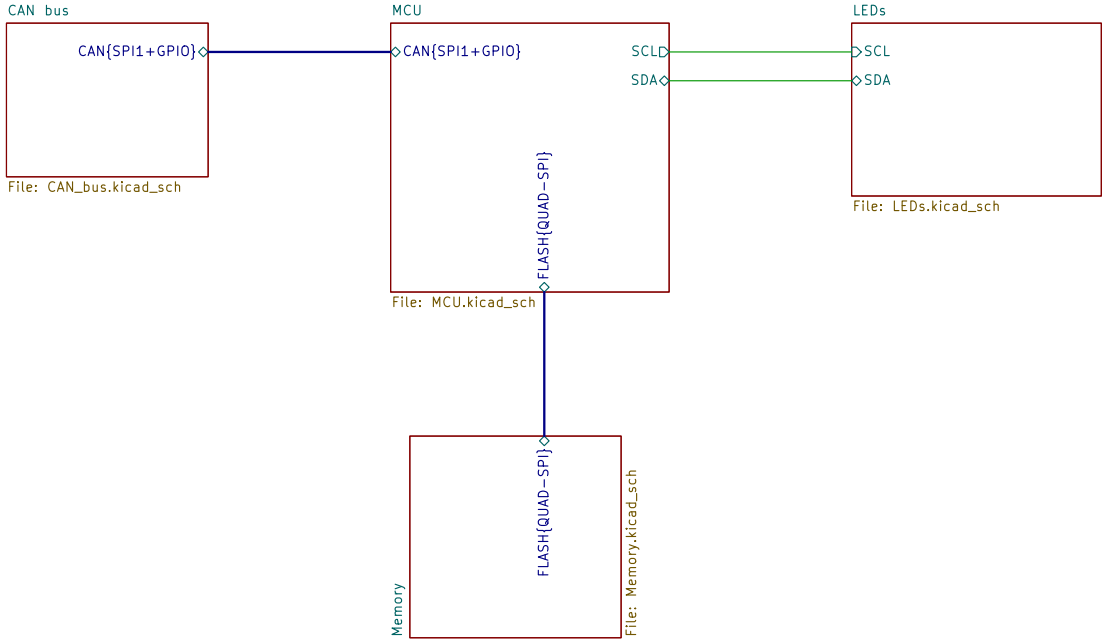


H101 MountingHole

H102 MountingHole

H103 MountingHole

H104 MountingHole

H105 MountingHole

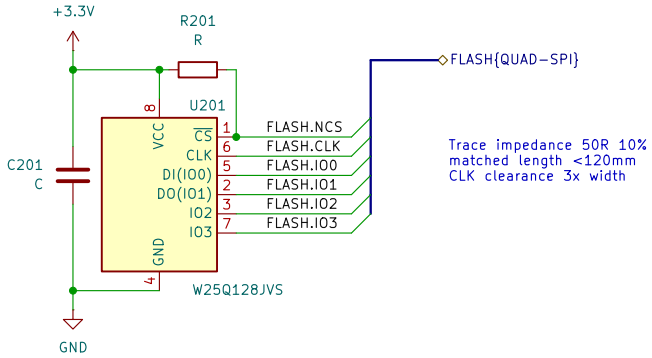
RnD Unit
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Sheet: /
File: BB_lighting_V1.kicad_sch

Title: BB_lighting

Size: A4	Date: 2024-03-21	**Rev: v1**
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Quad-SPI FLASH NOR Memory

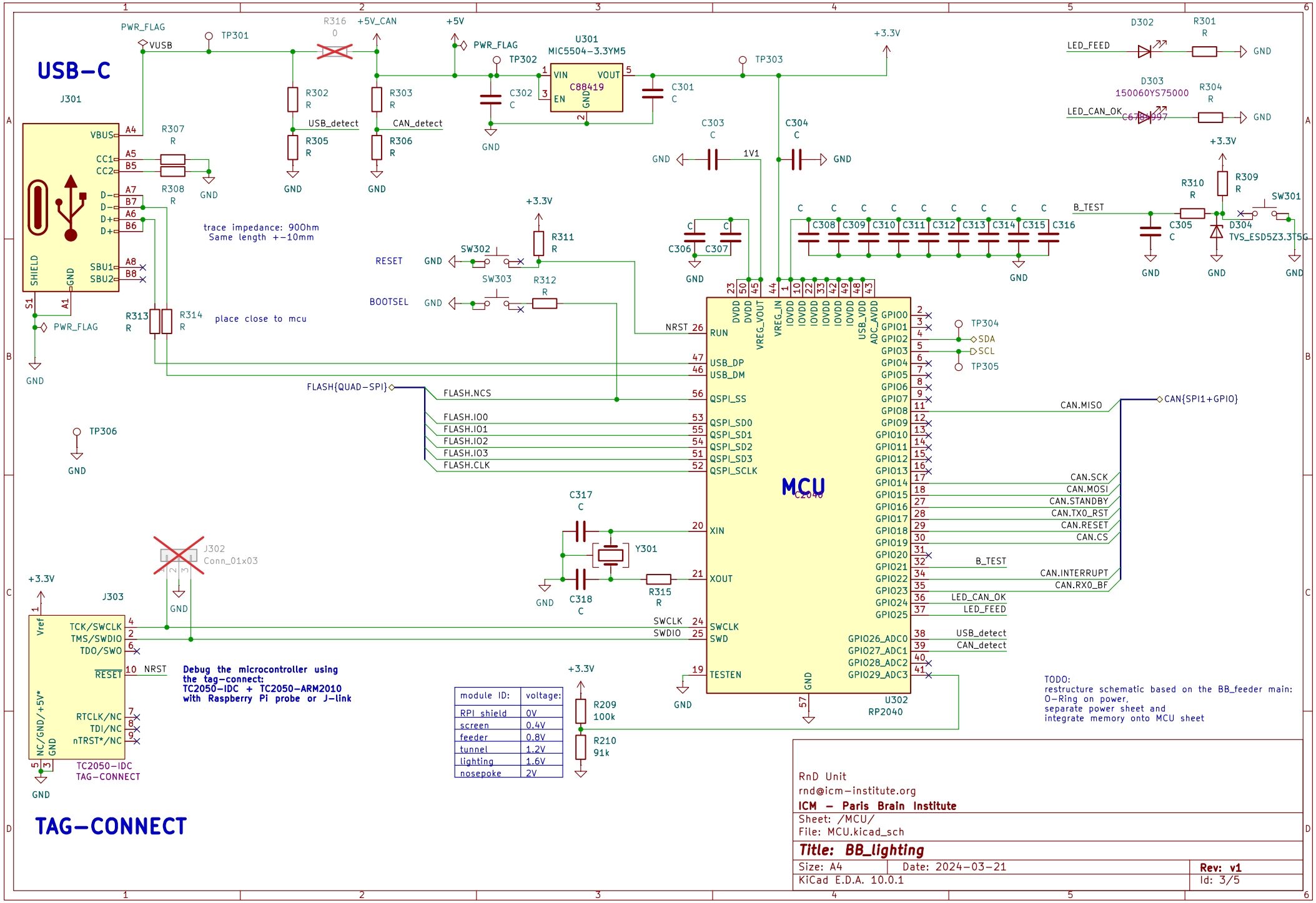


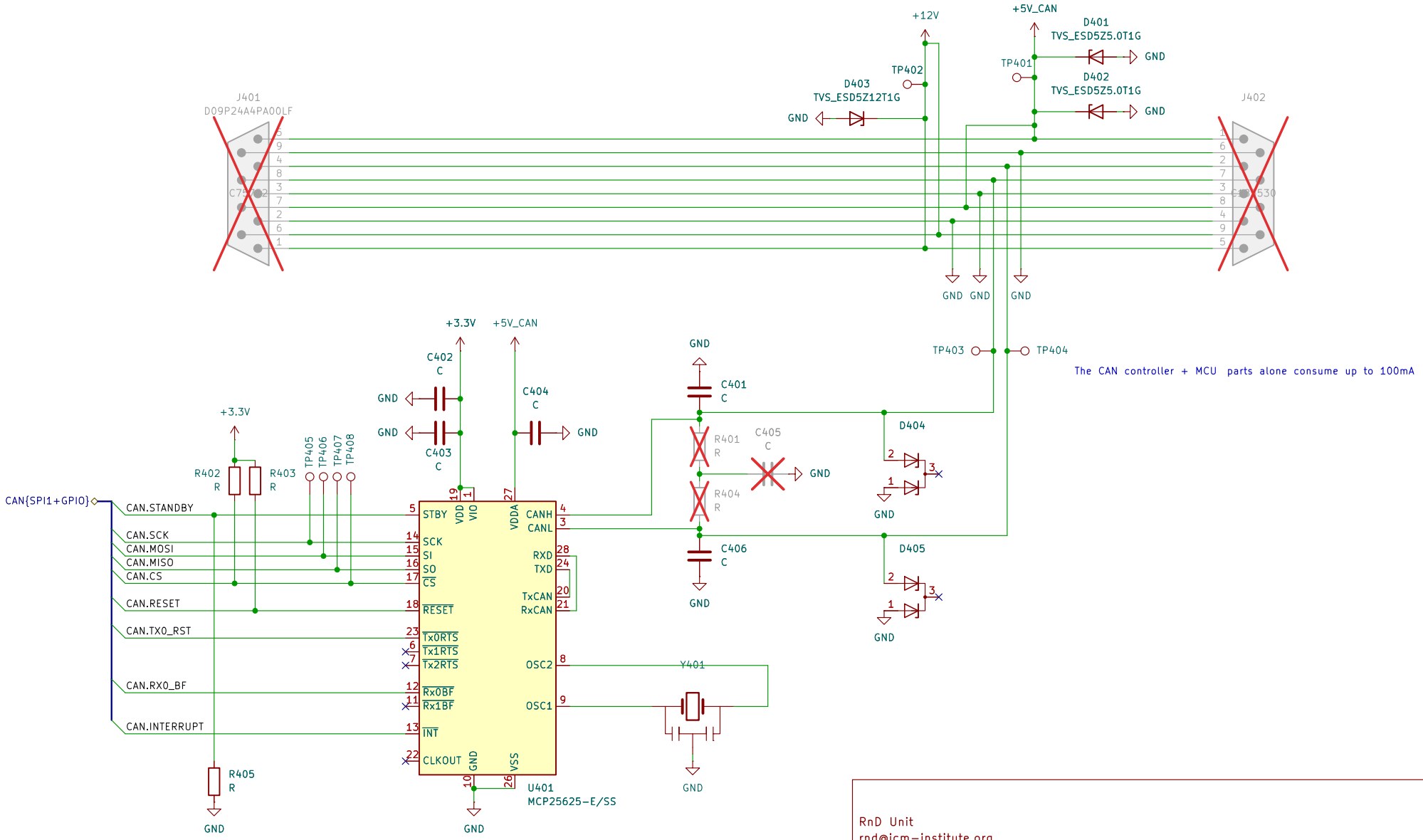
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File: Memory.kicad_sch

Title: BB_lighting

Size: A4	Date: 2024-03-21	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 2/5


USB-C
J301
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
A4
A5
B5
A7
B7
A6
B6
A8
B8
S1
A1
PWR_FLAG
VUSB
TP301
R316
0
+5V_CAN
+5V
PWR_FLAG
TP302
U301
MIC5504-3.3YM5
C88419
VIN
VOUT
EN
GND
C302
C
C301
C
TP303
+3.3V
R302
R
R303
R
USB_detect
CAN_detect
R305
R
R306
R
R307
R
R308
R
trace impedance: 90Ohm
Same length +-10mm
C303
C
C304
C
1V1
D302
R301
LED_FEED
D303
150060YS75000
R304
LED_CAN_OK
C6786997
R310
R
R309
R
SW301
B_TEST
C305
C
D304
TVS_ESD5Z3.3T5G
C306
C307
C308
C309
C310
C311
C312
C313
C314
C315
C316
SW302
RESET
SW303
BOOTSEL
R311
R
R312
R
R313
R
R314
R
place close to mcu
MCU
RP2040
U302
NRST
RUN
USB_DP
USB_DM
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
XIN
XOUT
SWCLK
SWD
TESTEN
DVDD
VREG_VOUT
VREG_IN
IOVDD
USB_VDD
ADC_AVDD
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
TP304
SDA
SCL
TP305
FLASH{QUAD-SPI}
FLASH.NCS
FLASH.IO0
FLASH.IO1
FLASH.IO2
FLASH.IO3
FLASH.CLK
TP306
C317
C
Y301
C318
C
R315
R
CAN.MISO
CAN.SCK
CAN.MOSI
CAN.STANDBY
CAN.TX0_RST
CAN.RESET
CAN.CS
CAN.INTERRUPT
CAN.RX0_BF
CAN{SPI1+GPIO}
LED_CAN_OK
LED_FEED
USB_detect
CAN_detect
J302
Conn_01x03
J303
Vref
TCK/SWCLK
TMS/SWDIO
TDO/SWO
RESET
RTCLK/NC
TDI/NC
nTRST*/NC
NC/GND/+5V*
GND
TC2050-IDC
TAG-CONNECT
TAG-CONNECT
Debug the microcontroller using
the tag-connect:
TC2050-IDC + TC2050-ARM2010
with Raspberry Pi probe or J-link
module ID:
voltage:
RPI shield
0V
screen
0.4V
feeder
0.8V
tunnel
1.2V
lighting
1.6V
nosepoke
2V
R209
100k
R210
91k
TODO:
restructure schematic based on the BB_feeder main:
O-Ring on power,
separate power sheet and
integrate memory onto MCU sheet
RnD Unit
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Sheet: /MCU/
File: MCU.kicad_sch
Title: BB_lighting
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 10.0.1
Id: 3/5

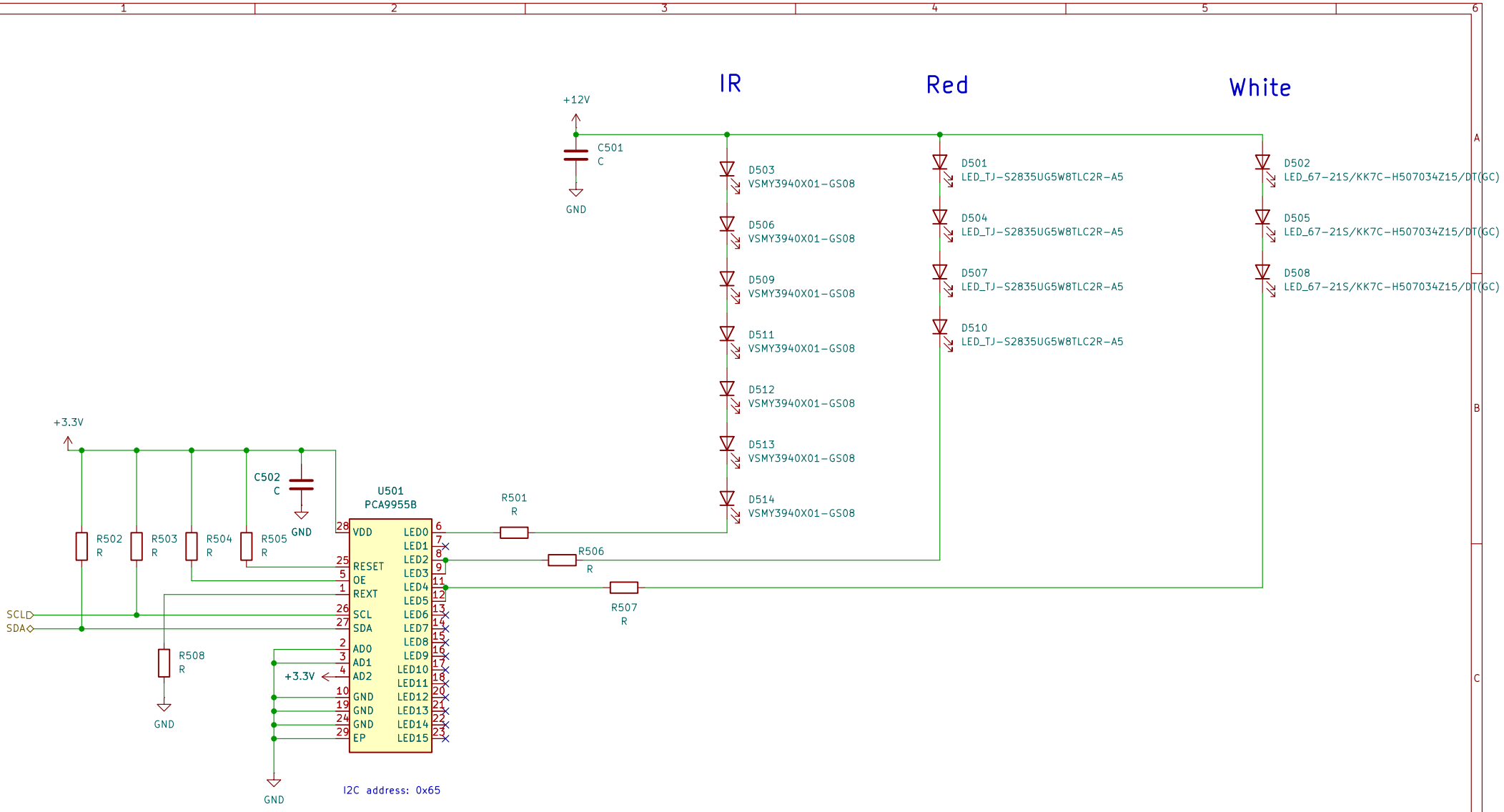


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Size: A4	Date: 2024-03-21	**Rev: v1**
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